LAB ASSIGNMENT-20.3

NAME:P.VIKAS

BATCH:05

ROLL NO. 2403A510E6

Task 1 – Input Validation Check

Prompt

Prompt AI to generate a simple Python username-password login program. Then review whether input sanitization is done. Improve the script using regular expressions for validation.

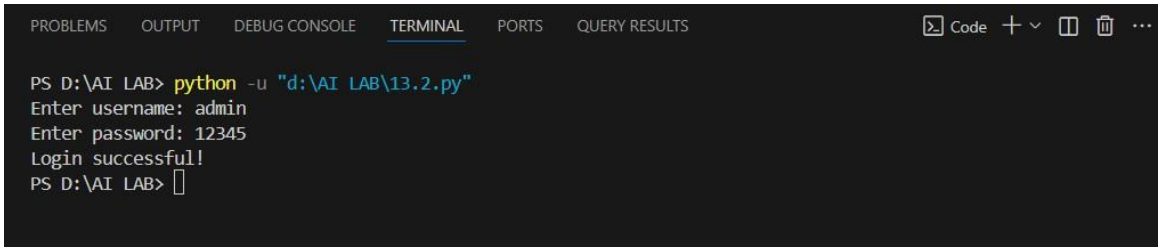
Insecure Code

```
# Insecure login script (AI-generated)

username = input("Enter username: ")
password = input("Enter password: ")

if username == "admin" and password == "12345":
    print("Login successful!")
else:
    print("Invalid credentials.")
```

Output



Observation

+ No input validation. + Password stored in plaintext. + Susceptible to brute force and injection-style attacks.

Observation

■ Secure input validation using regex ■ Hides password input ■ Prevents invalid or malicious inputs

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Task 2 – SQL Injection Prevention

Prompt

Ask AI to generate a Python script that fetches user details using SQLite. Identify if it's vulnerable to SQL injection, then refactor using parameterized queries.

Code

```
import sqlite3


def create_sample_database():
    """Create a sample users database with test data"""
    try:
        conn = sqlite3.connect("users.db")
        cursor = conn.cursor()

        # Drop table if it exists (for fresh start)
        cursor.execute("DROP TABLE IF EXISTS users")

        # Create users table
        cursor.execute("""
            CREATE TABLE users (
                id INTEGER PRIMARY KEY,
                username TEXT NOT NULL,
                email TEXT NOT NULL,
                age INTEGER
            )
        """)

        # Insert sample data
        sample_users = [
            ('john_doe', 'john@example.com', 28),
            ('jane_smith', 'jane@example.com', 25),
            ('bob_johnson', 'bob@example.com', 35),
            ('alice_brown', 'alice@example.com', 30),
            ('charlie_davis', 'charlie@example.com', 22)
        ]

        cursor.executemany("""
            INSERT INTO users (username, email, age) VALUES (?, ?, ?)
        """, sample_users)

        conn.commit()
        print("✓ Sample database created with test data\n")
```

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```
        conn.close()

    except sqlite3.Error as e:
        print(f"Database creation error: {e}")

def search_user():
    """Search for a user by username"""
    try:
        conn = sqlite3.connect("users.db")
        cursor = conn.cursor()

        username = input("Enter username to search: ")

        # Use parameterized query to prevent SQL injection
        query = "SELECT * FROM users WHERE username = ?"
        cursor.execute(query, (username,))

        results = cursor.fetchall()

        if results:
            print("\n✓ User found:")
            print("-" * 50)
            for row in results:
                print(f"ID: {row[0]}")
                print(f"Username: {row[1]}")
                print(f"Email: {row[2]}")
                print(f"Age: {row[3]}")
            print("-" * 50)
        else:
            print(f"\n✗ No user found with username: {username}")

    except sqlite3.Error as e:
        print(f"Database error: {e}")
    except Exception as e:
        print(f"Error: {e}")
    finally:
        if conn:
            conn.close()

def display_all_users():
    """Display all users in the database"""
    try:
        conn = sqlite3.connect("users.db")
        cursor = conn.cursor()
```

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```
        cursor.execute("SELECT * FROM users")
        results = cursor.fetchall()

        print("\n" + "="*60)
        print("ALL USERS IN DATABASE")
        print("="*60)
        for row in results:
            print(f"ID: {row[0]:2} | Username: {row[1]:15} | Email:
{row[2]:20} | Age: {row[3]}")
        print("="*60 + "\n")

        conn.close()
    except sqlite3.Error as e:
        print(f"Database error: {e}")

# Main program
if __name__ == "__main__":
    print("╔═══════════════════════════════════════════╗")
    print("║   USER DATABASE SEARCH APPLICATION        ║")
    print("╚═══════════════════════════════════════════╝\n")

    # Create sample database
    create_sample_database()

    # Display all users
    display_all_users()

    # Search for a user
    search_user()
```

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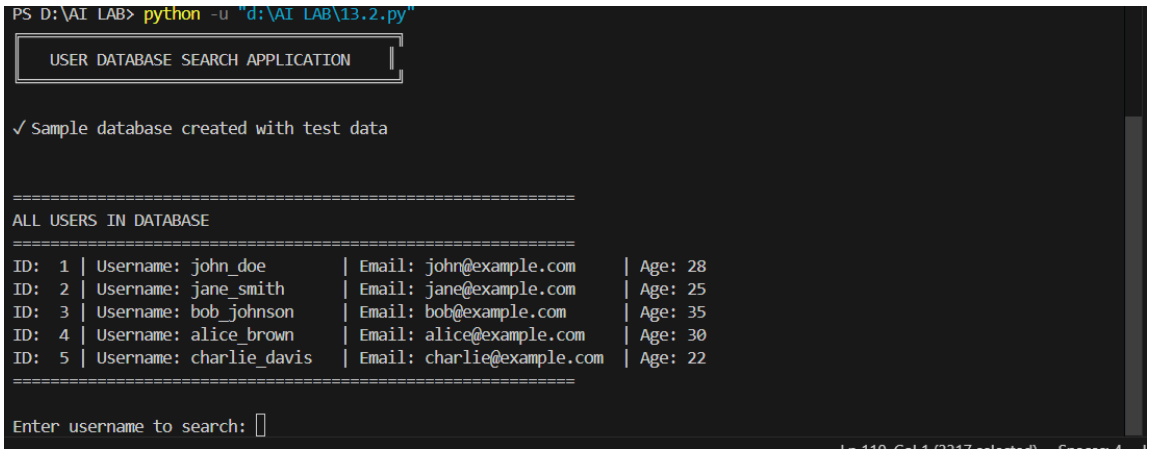
Output :



Observation

ı Vulnerable to SQL Injection due to direct string concatenation.

■ Uses prepared statements ■ Eliminates SQL injection risk ■ Works for valid usernames safely

Task 3 – Cross-Site Scripting (XSS) Check

Prompt

Generate an HTML feedback form with JavaScript that displays user feedback. Check for XSS vulnerability and fix it by escaping input.

Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Feedback Form</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            min-height: 100vh;
```

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```
        display: flex;
        justify-content: center;
        align-items: center;
        padding: 20px;
    }

    .container {
        background: white;
        padding: 40px;
        border-radius: 10px;
        box-shadow: 0 10px 25px rgba(0, 0, 0, 0.2);
        max-width: 500px;
        width: 100%;
    }

    h2 {
        color: #333;
        margin-bottom: 30px;
        text-align: center;
    }

    .form-group {
        margin-bottom: 20px;
    }

    textarea {
        width: 100%;
        padding: 12px;
        border: 2px solid #ddd;
        border-radius: 5px;
        font-size: 14px;
        font-family: Arial, sans-serif;
        resize: vertical;
        min-height:  120px;
        transition: border-color 0.3s;
    }

    textarea:focus {
        outline: none;
        border-color: #667eea;
    }

    button {
        width: 100%;
        padding: 12px;
```

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```
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        color: white;
        border: none;
        border-radius: 5px;
        font-size: 16px;
        font-weight: bold;
        cursor: pointer;
        transition: transform 0.2s;
    }

    button:hover {
        transform: translateY(-2px);
        box-shadow: 0 5px 15px rgba(102, 126, 234, 0.4);
    }

    button:active {
        transform: translateY(0);
    }

    #output {
        margin-top: 20px;
        padding: 15px;
        background: #f0f0f0;
        border-left: 4px solid #667eea;
        border-radius: 5px;
        color: #333;
        display: none;
        animation: slideIn 0.3s ease;
    }

    #output.show {
        display: block;
    }

    @keyframes slideIn {
        from {
            opacity: 0;
            transform: translateY(-10px);
        }
        to {
            opacity: 1;
            transform: translateY(0);
        }
    }
```

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```
        .success {
            color: #27ae60;
            font-weight: bold;
        }
    </style>
</head>
<body>
    <div class="container">
        <h2>📝 Feedback Form</h2>

        <div class="form-group">
            <textarea id="feedback" placeholder="Enter your feedback
here..."></textarea>
        </div>

        <button onclick="submitFeedback()">Submit Feedback</button>

        <div id="output"></div>
    </div>

    <script>
        function submitFeedback() {
            const feedback =
document.getElementById("feedback").value.trim();
            const outputDiv = document.getElementById("output");

            // Validate input
            if (feedback === "") {
                outputDiv.innerHTML = '<span style="color: #e74c3c;">❌
Please enter feedback before submitting!</span>';
                outputDiv.classList.add("show");
                return;
            }

            // Display success message
            outputDiv.innerHTML = '<span class="success">✓ Thank you for
your feedback!</span><br><br><strong>Your feedback:</strong><br>' +
escapeHtml(feedback);
            outputDiv.classList.add("show");

            // Clear the textarea
            document.getElementById("feedback").value = "";

            // Optional: Fade out after 5 seconds
            setTimeout(() => {
```

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```
                outputDiv.classList.remove("show");
            }, 5000);
        }

        // Prevent XSS attacks by escaping HTML
        function escapeHtml(text) {
            const map = {
                '&': '&amp;',
                '<': '&lt;',
                '>': '&gt;',
                '"': '&quot;',
                "'": '&#039;'
            };
            return text.replace(/[&<>"']/g, m => map[m]);
        }

        // Allow Enter key to submit feedback
        document.getElementById("feedback").addEventListener("keydown",
function(event) {
            if (event.key === "Enter" && event.ctrlKey) {
                submitFeedback();
            }
        });
    </script>
</body>
</html>
```

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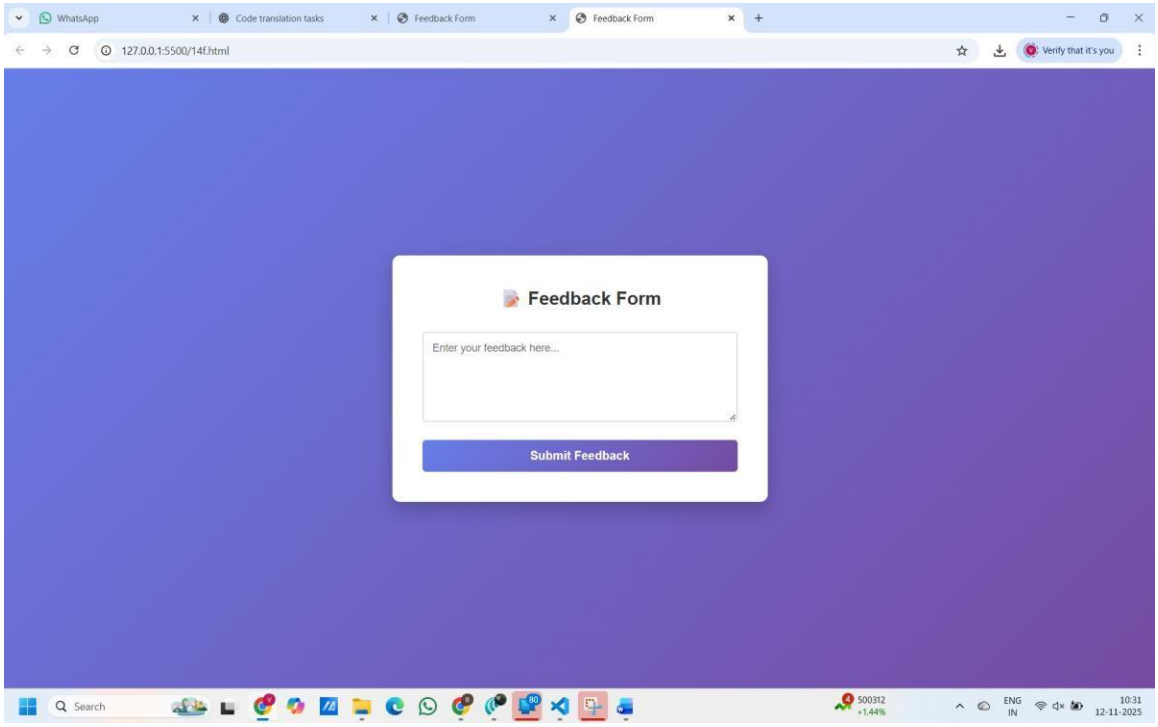
Output



Observation

■ Prevents execution of injected scripts ■ Uses HTML escaping to mitigate XSS ■ Safe rendering of untrusted input

Task 4 – Real-Time Application: Security Audit

Prompt

Select an AI-generated login form with file upload and perform a security audit. Identify vulnerabilities and rewrite securely.

Code:

```
from flask import Flask, request, jsonify, render_template_string

from werkzeug.utils import secure_filename
import os

app = Flask(__name__)
UPLOAD_FOLDER = "safe_uploads"
ALLOWED_EXTENSIONS = {'png', 'jpg', 'txt'}
MAX_FILE_SIZE = 5 * 1024 * 1024  # 5MB limit

# Create upload folder if it doesn't exist
if not os.path.exists(UPLOAD_FOLDER):
    os.makedirs(UPLOAD_FOLDER)

app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER
```

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```
app.config['MAX_CONTENT_LENGTH'] = MAX_FILE_SIZE

def allowed_file(filename):
    """Check if file extension is allowed"""
    return '.' in filename and filename.rsplit('.', 1)[1].lower() in
ALLOWED_EXTENSIONS

@app.route('/', methods=['GET'])
def index():
    """Display the upload form"""
    html = '''
    <!DOCTYPE html>
    <html>
    <head>
        <title>File Upload</title>
        <style>
            body {
                font-family: Arial, sans-serif;
                max-width: 600px;
                margin: 50px auto;
                padding: 20px;
                background: #f5f5f5;
            }
            .container {
                background: white;
                padding: 30px;
                border-radius: 8px;
                box-shadow: 0 2px 10px rgba(0,0,0,0.1);
            }
            h1 { color: #333; }
            input[type="file"] {
                padding: 10px;
                margin: 10px 0;
            }
            button {
                background: #4CAF50;
                color: white;
                padding: 10px 20px;
                border: none;
                border-radius: 4px;
                cursor: pointer;
                font-size: 16px;
            }
            button:hover { background: #45a049; }
```

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```
            .message { margin-top: 20px; padding: 10px; border-radius:
4px; }
            .success { background: #d4edda; color: #155724; }
            .error { background: #f8d7da; color: #721c24; }
        </style>
    </head>
    <body>
        <div class="container">
            <h1>🗁 File Upload</h1>
            <p>Allowed file types: PNG, JPG, TXT (Max 5MB)</p>
            <form method="POST" enctype="multipart/form-data">
                <input type="file" name="file" required>
                <button type="submit">Upload</button>
            </form>
            <div id="message"></div>
        </div>
    </body>
    </html>
    '''
    return render_template_string(html)

@app.route('/upload', methods=['POST'])
def upload_file():
    """Handle file upload"""
    try:
        # Check if file is in request
        if 'file' not in request.files:
            return jsonify({"error": "No file part"}), 400

        file = request.files['file']

        # Check if file is selected
        if file.filename == '':
            return jsonify({"error": "No selected file"}), 400

        # Validate file
        if not allowed_file(file.filename):
            return jsonify({"error": f"Invalid file type. Allowed: {',
'.join(ALLOWED_EXTENSIONS)}"}), 400

        # Check file size
        file.seek(0, os.SEEK_END)
        file_length = file.tell()
        if file_length > MAX_FILE_SIZE:
```

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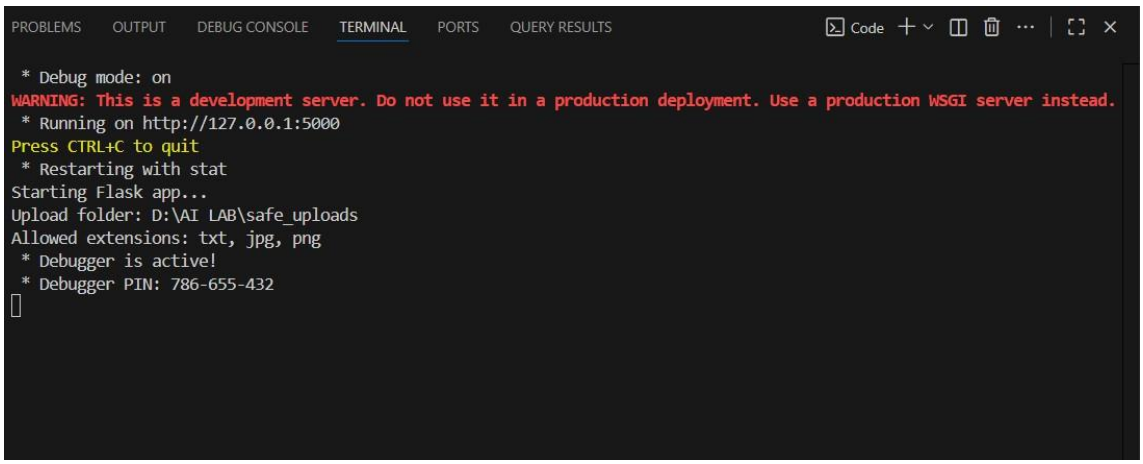
```
            return jsonify({"error": f"File too large. Max size: 5MB"}),
400
        file.seek(0)

        # Save file securely
        filename = secure_filename(file.filename)
        path = os.path.join(app.config['UPLOAD_FOLDER'], filename)
        file.save(path)

        return jsonify({"success": f"File '{filename}' uploaded
successfully!", "path": path}), 200

    except Exception as e:
        return jsonify({"error": f"Upload failed: {str(e)}"}), 500

@app.route('/files', methods=['GET'])
def list_files():
    """List all uploaded files"""
    try:
        files = os.listdir(UPLOAD_FOLDER)
        return jsonify({"files": files, "count": len(files)}), 200
    except Exception as e:
        return jsonify({"error": str(e)}), 500

if __name__ == "__main__":
    print("Starting Flask app...")
    print(f"Upload folder: {os.path.abspath(UPLOAD_FOLDER)}")
    print(f"Allowed extensions: {', '.join(ALLOWED_EXTENSIONS)}")
    app.run(debug=True, port=5000)
```

output:



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Observation

■ Safe upload path ■ Restricted extensions ■ Prevents path traversal & malicious uploads